

Data Wrangling – Problem Solving (SQL Notes)

(i) Show all customers living in California

First guess

```
SELECT *  
FROM mski_customers  
WHERE state = 'california';
```

This is almost right but not the proper one

Why?

Because of **data quality issue**.

Let's check

```
SELECT DISTINCT state  
FROM mski_customers;
```

We can see:

- ' California'
- 'california '
- ' california'
- 'california'

👉 Same value but extra spaces.

So now what is the proper query?

Which function should we use?

What is LIKE operator?

Used in WHERE clause to search for a specific pattern.

```
SELECT *  
FROM mski_customers  
WHERE state LIKE 'california';
```

Still not enough. Something missing ❌

Final correct query

```
SELECT *  
FROM mski_customers  
WHERE TRIM(state) LIKE 'california';
```

TRIM() removes extra white spaces before comparison.

(ii) All customers living in Florida or New York & belonging to Consumer segment

Let me show you data:

```
SELECT *
```

```
FROM mski_customers;
```

First attempt

```
SELECT *
```

```
FROM mski_customers
```

```
WHERE state = 'florida' OR state = 'new york'
```

```
AND segment = 'consumer';
```

✗ This is wrong because AND has higher priority than OR

First way (using brackets)

```
SELECT *
```

```
FROM mski_customers
```

```
WHERE (state = 'florida' OR state = 'new york')
```

```
AND segment = 'consumer';
```

Second way (using IN)

```
SELECT *
```

```
FROM mski_customers
```

```
WHERE state IN ('florida','new york')
```

```
AND segment = 'consumer';
```

(iii) Print customer names in the following format

Rob Lucas → R. Lucas

Let me explain using example:

ASHISH SHAH → A. SHAH

Logic

- Take first letter of first name

- Find space position
- Extract last name

Final query

```
SELECT
LEFT(customer_name, 1) + ' ' +
SUBSTRING(
    customer_name,
    CHARINDEX(' ', customer_name) + 1,
    LEN(customer_name) - CHARINDEX(' ', customer_name)
) AS formatted_name
FROM mski_customers;
```

(iv) How many customers are having an outlook.com email

From customers table:

```
SELECT *
FROM customers;
```

Query

```
SELECT *
FROM customers
WHERE customer_email LIKE '%@outlook.com';
```

If you want only the count

```
SELECT COUNT(*)
FROM customers
WHERE customer_email LIKE '%@outlook.com';
```

(v) Orders placed in 2018

```
.SELECT *
FROM orders;
We choose order_purchase_date.
```

a. SELECT COUNT(DISTINCT customer_id)

```
FROM orders  
WHERE YEAR(order_purchase_date) = 2018;
```

b

```
SELECT COUNT(DISTINCT customer_id)  
FROM orders  
WHERE DATEPART(year, order_purchase_date) = 2018;
```

c

```
SELECT COUNT(DISTINCT customer_id)  
FROM orders  
WHERE DATENAME(year, order_purchase_date) = '2018';
```

(vi) Date functions

1. YEAR()
2. DATEPART()
3. DATENAME()

Difference will be explained in next question.

(vii) How many customers in Dec 2017

Using YEAR and MONTH

```
SELECT COUNT(DISTINCT customer_id)  
FROM orders  
WHERE YEAR(order_purchase_date) = 2017  
AND MONTH(order_purchase_date) = 12;
```

Using DATEPART

```
SELECT COUNT(DISTINCT customer_id)  
FROM orders  
WHERE DATEPART(year, order_purchase_date) = 2017
```

AND DATEPART(month, order_purchase_date) = 12;

Using DATENAME

SELECT COUNT(DISTINCT customer_id)

FROM orders

WHERE DATENAME(year, order_purchase_date) = '2017'

AND DATENAME(month, order_purchase_date) = 'December';

Note:

- DATEPART → integer
- DATENAME → string

(viii) With 24 hours

SELECT *

FROM orders;

Which columns are we going to use:

- order_purchase_date
- order_approved_at

Correct query

SELECT COUNT(*)

FROM orders

WHERE DATEDIFF(

hour,

order_purchase_date,

order_approved_at

) <= 24;

Why not DAY?

DATEDIFF(day, order_purchase_date, order_approved_at) <= 1

✗ Not same because day difference ignores hours.

(ix)

SELECT COUNT(*)

FROM orders

WHERE DATEPART(weekday, order_estimated_delivery_date) > 1

AND DATEPART(weekday, order_estimated_delivery_date) < 7;

OR using BETWEEN

SELECT COUNT(*)

FROM orders

WHERE DATEPART(weekday, order_estimated_delivery_date)

BETWEEN 2 AND 6;

(x) Using CASE

SELECT *,

CASE

 WHEN profit > 0 THEN 'profit'

 ELSE 'loss'

END AS status

FROM transactions;

(xi)

Try this on your own.